

On the Measure of Intelligence

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Feliciann Elliot

MAI 5301 - Foundations Of Large
Language Models

Abstract

To make real progress toward human-like AI, we need a clear way to define and measure intelligence so we can fairly compare systems to each other and to humans. Historically, both psychology and AI have proposed many definitions, but modern AI still tends to measure intelligence through task performance (e.g., games). The problem is that task skill can be boosted by massive training data or prior knowledge, which hides true generalization ability. The authors instead define intelligence as skill-acquisition efficiency, grounded in Algorithmic Information Theory, emphasizing scope, priors, experience, and generalization difficulty. Based on this, they propose guidelines for proper general AI benchmarks and introduce ARC (Abstraction and Reasoning Corpus), designed to measure human-like general fluid intelligence under controlled priors.

Introduction

Artificial intelligence research has achieved remarkable benchmark performance across vision, language, and reasoning tasks. However, high task performance does not necessarily imply general intelligence. Many systems achieve success by leveraging massive training data, task-specific optimization, and accumulated priors rather than demonstrating true adaptability. This paper addresses a fundamental question: how should intelligence be defined and measured in a way that distinguishes generalization ability from mere skill accumulation? By reframing intelligence as the efficiency with which a system converts prior knowledge and experience into new skills, the paper challenges conventional benchmark-based evaluation and proposes a more principled framework for measuring general intelligence.

What Problem Is This Paper Solving?

- AI progress is often measured by task skill, not general intelligence
- Skill can be “bought” with more data and more priors (hard-coding, pretraining, task leakage)

Without a clear definition/metric, evaluation becomes an implicit, inconsistent target

What This Paper Calls “Intelligence”

- Intelligence is not defined as skill level at a known task, but rather as the efficiency of skill acquisition under uncertainty.
- The high-level objective is to quantify generalization power across novel situations rather than measuring performance on familiar problems alone.

Why “Skill Benchmarks” Fail as Intelligence Tests

If the task is known in advance, AI systems can be optimized via:

- A Knowledge Base
- Training data

This produces high scores with weak generalization.

Hence, benchmark scores may reflect an engineering investment, not intelligence

How ARC Differs from Traditional Benchmarks

- ImageNet: 14M+ labeled images, large-scale supervised training, measures pattern recognition.
- GLUE: Fixed NLP datasets, task-specific training, measures benchmark performance.
- ARC: ~3 examples per task, no task-specific training allowed, measures abstraction and generalization efficiency.
- Key Purpose: Isolate skill-acquisition efficiency rather than accumulated skill.

Key Terms of the Field

Priors (P): information the system starts with (architecture, pretraining, hand-coded knowledge)

Experience (E): information gained during training/interaction

Generalization difficulty (GD): uncertainty about future situations beyond observed data

Scope: the task distribution where “intelligence” is being claimed

Key Terms of the Field

Intelligence is the rate a learner converts priors + experience → skill

Intelligence must be evaluated across tasks with real uncertainty

Skill is an output artifact; intelligence is the underlying adaptation process.

Why "Universal Intelligence" is Rejected

- Intelligence cannot be meaningfully measured on an absolute universal scale.
- Valid comparisons require shared task scope and comparable prior knowledge between systems being evaluated.

The paper argues an anthropocentric frame is necessary for actionable measurement

A Baseline for Fair Comparison

Humans are not blank slates; we possess evolved priors that enable data-efficient learning from limited experience. The paper adopts Core Knowledge theory as a plausible characterization of innate human cognitive primitives.

- Fair intelligence tests should explicitly control for prior knowledge rather than ignoring its existence.
- Meaningful comparisons require either matching priors between systems or accounting for prior differences when interpreting performance gaps.

Core Knowledge Priors Used as Benchmark Assumptions

The test assumes humans possess the following innate cognitive primitives:

- Agentness and goal-directedness (understanding intentional behavior)
- Natural numbers and basic arithmetic operations
- Elementary geometry and topological relationships

Tests should assume only these foundational priors, excluding learned cultural facts or domain-specific knowledge acquired through education or experience.

The Proposed Measure

Task contribution is conceptually:

- $\text{Skill} \times \text{generalization_difficulty} / (\text{priors} + \text{experience})$
- Weighted by task value to compare across tasks
- Intelligence is averaged over tasks and curricula

ARC: Benchmark Designed From the Definition

- The paper proposes ARC (Abstraction and Reasoning Corpus) as an actionable benchmark aligned with the definition
- The goal is to measure human-like fluid intelligence via abstraction and reasoning
- It is designed to minimize dependence on an acquired domain knowledge

How ARC Tasks Were Designed

Task Architecture

- Format: Grid-based puzzles (1×1 to 30×30) using 10 color-coded symbols.
- Few-Shot Learning: Average of 3.3 examples to infer a transformation rule.
- Program Synthesis: No multiple choice; the agent must construct the output grid from scratch (size, color, and placement).

How ARC Tasks Were Designed

The "Core Knowledge" Priors:

- ARC assumes only innate human cognitive abilities, avoiding "crystallized" or learned knowledge.
- Objectness: Identifying distinct shapes, persistence, and physical contact.
- Goal-Directedness: Tracking intentionality or "agents" moving toward targets.
- Arithmetic: Counting, sorting, and basic operations for small quantities (<10). Geometry/Topology: Symmetries, rotations, scaling, and "inside vs. outside."

How ARC Tasks Were Designed

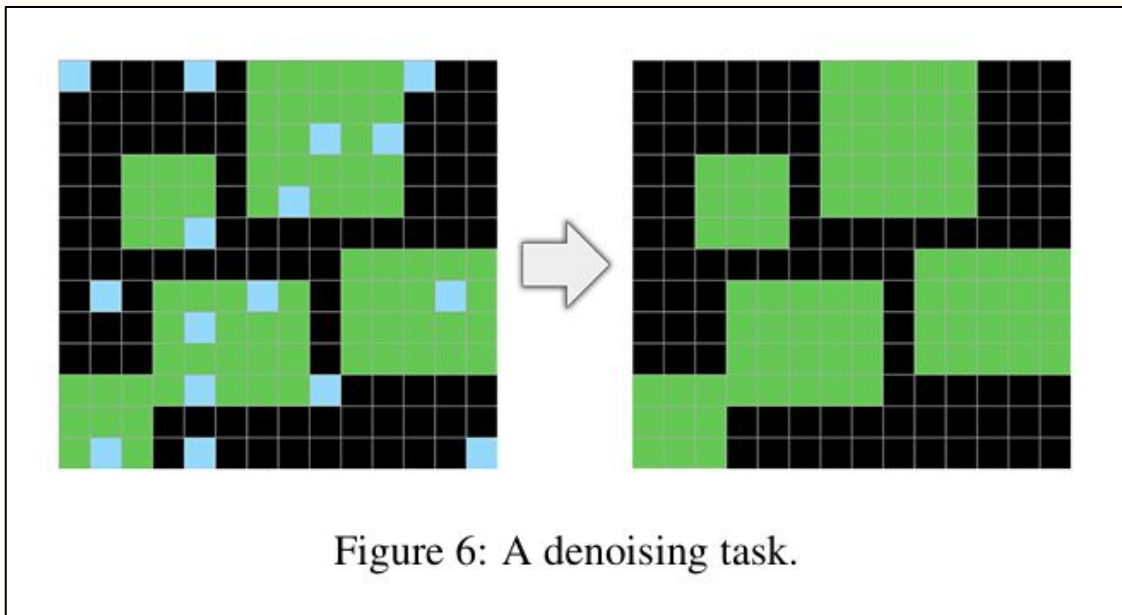
Evaluation Rigor

- Manual Generation: Tasks are hand-crafted to ensure high diversity and prevent "cheating" via reverse-engineering generative code.
- Developer-Aware Generalization: Evaluation tasks are entirely novel and "unknown unknowns," specifically testing fluid intelligence over memorization.

Why Manual Task Design Matters

- Tasks are hand-designed, not algorithmically generated.
- Prevents reverse-engineering of task generation patterns.
- Avoids dataset leakage and shortcut exploitation.
- Forces genuine reasoning instead of memorization.

Example ARC Task 1: Object Segmentation



(Chollet, 2019) - A denoising task

Task requires identifying coherent object structure within noisy grid

Must infer rule: remove extraneous pixels while preserving dominant pattern

Tests:

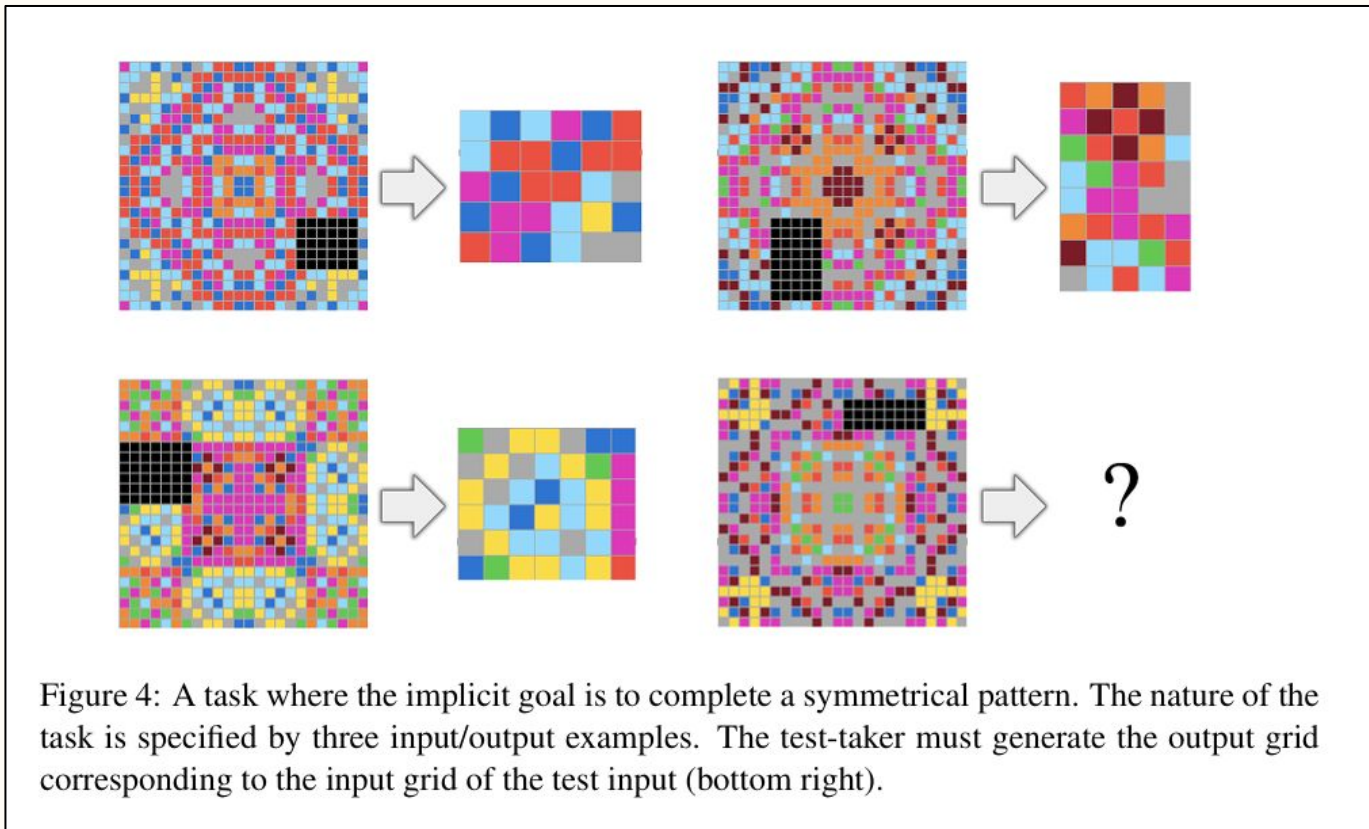
- Object boundary detection
- Signal-noise discrimination
- Invariant structural preservation

No statistical pattern matching possible (grid too small)

Concrete ARC Example - Arithmetic/Counting

- Example Rule: Keep only the color that appears most frequently in the grid.
- Model must infer counting, grouping, and majority concept from only 3 examples.
- Tests small-number arithmetic (<10), object grouping, abstraction, and generalization.
- No statistical shortcut possible due to small, unique grids.

Example ARC Task 2: Symmetry Completion



ARC - Compositional Reasoning Example

- Example Rule: Identify object $\rightarrow$ Rotate 90° $\rightarrow$ Duplicate $\rightarrow$ Mirror horizontally.
- System must infer multiple primitive operations and compose them correctly.
- Tests program synthesis, compositional reasoning, structural abstraction.
- Each task appears once; memorization is impossible.

What These ARC Examples Demonstrate

Tasks require inferring geometric or logical rules from minimal examples and generalizing to unseen configurations. This structure demonstrates three core cognitive capabilities:

- Compositional reasoning: combining primitive operations into complex transformations
- Structural abstraction: identifying underlying patterns beyond surface features
- Generalization beyond memorization: applying learned rules to novel instances rather than retrieving similar training examples

ARC: Benchmark Designed From the Definition

ARC (Abstraction and Reasoning Corpus) can be interpreted through three complementary lenses: as a general artificial intelligence benchmark, a program synthesis challenge, or a psychometric intelligence test.

The benchmark targets both humans and AI systems aiming to emulate human-like general fluid intelligence, the ability to reason abstractly and solve novel problems without relying on prior knowledge.

Human vs AI Results on ARC

- Humans: ~85% average accuracy.
- Early deep learning systems: near 0%.
- Specialized ARC solvers: still far below human performance.
- Interpretation: Scaling data alone does not produce systematic generalization.

What Makes ARC Different ?

- No massive training set per task (This reduces “buying skill with data”)
- Emphasizes few-shot abstraction, compositional generalization
- Attempts to isolate generalization strength rather than memorization

Technical Impact on Large Language Model Evaluation

- No massive training set per task (This reduces “buying skill with data”)
- Emphasizes few-shot abstraction, compositional generalization
- Attempts to isolate generalization strength rather than memorization

Limitations

- “Priors” are hard to quantify for LLMs (pretraining datasets constitute enormous implicit prior knowledge)
- Task value assignments are inherently subjective and directly affect intelligence comparisons. Different weighting schemes can produce contradictory rankings of the same systems.
- The Anthropocentric scope is practical, but excludes non-human-relevant intelligence forms

Conclusion

- Intelligence should not be equated with benchmark scores; intelligence is fundamentally the efficiency of skill acquisition under novel conditions.
- Any serious intelligence metric must explicitly control for prior knowledge, experience quantity, and generalization difficulty to enable fair comparison.
- ARC (Abstraction and Reasoning Corpus) is proposed as a benchmark designed to align with these theoretical constraints, testing generalization while controlling for human-like priors.

References

Chollet, F. (2019, November 5). On the Measure of Intelligence. arXiv.org. <https://arxiv.org/abs/1911.01547>



Questions?

Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference

arXiv:2403.04132

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Abstract

Large Language Models (LLMs) have enabled powerful new applications, but evaluating how well they align with human preferences remains difficult. To tackle this, the authors introduce Chatbot Arena, an open platform that ranks LLMs using crowdsourced, pairwise human comparisons. After collecting over 240,000 votes, the platform applies established statistical methods to reliably evaluate and rank models. Results show that crowd-generated questions are diverse and discriminative, and that crowd judgments closely match expert evaluations. These findings support the credibility of Chatbot Arena, which has since become one of the most widely cited LLM leaderboards.

Introduction

Recent advances in LLMs have expanded their capabilities, but evaluating them especially on real-world, open-ended tasks and alignment with human preferences remains a challenge. Most existing benchmarks rely on static, ground-truth datasets because they are cheap and reproducible. However, these benchmarks are limited: they are not open-ended, can become contaminated over time, and often lack clear ground truth for complex tasks. As a result, they fail to properly assess modern LLMs, particularly in terms of user preference.

To address this, the authors introduce Chatbot Arena, a live, crowdsourced evaluation platform where users compare anonymous model responses and vote for the better one. This setup collects fresh, diverse prompts and reflects real-world usage. Using statistical ranking methods (e.g., Bradley–Terry models and E-values), the platform efficiently and reliably ranks models while handling noisy human feedback.

Since launching in April 2023, Chatbot Arena has gathered over 240,000 votes from around 90,000 users across 100+ languages, featuring more than 50 state-of-the-art models. Analyses show that the prompts are diverse and discriminative, and that crowd judgments align closely with expert evaluations. The platform has become one of the most referenced LLM leaderboards.

What Problem Is This Paper Solving?

- Traditional Natural Language Processing benchmarks are static and saturate quickly
- Models are often trained directly on benchmark-like data
- Automatic metrics (BLEU, ROUGE) poorly reflect real-world usefulness
- Evaluation lacks dynamic, large-scale human feedback

The Core Idea of Chatbot Arena

It is a public platform where users compare two anonymous model responses

- Users choose the better response (pairwise comparison)
- Models are ranked using Elo rating
- Continuous, open-ended evaluation

Chatbot Arena Interface

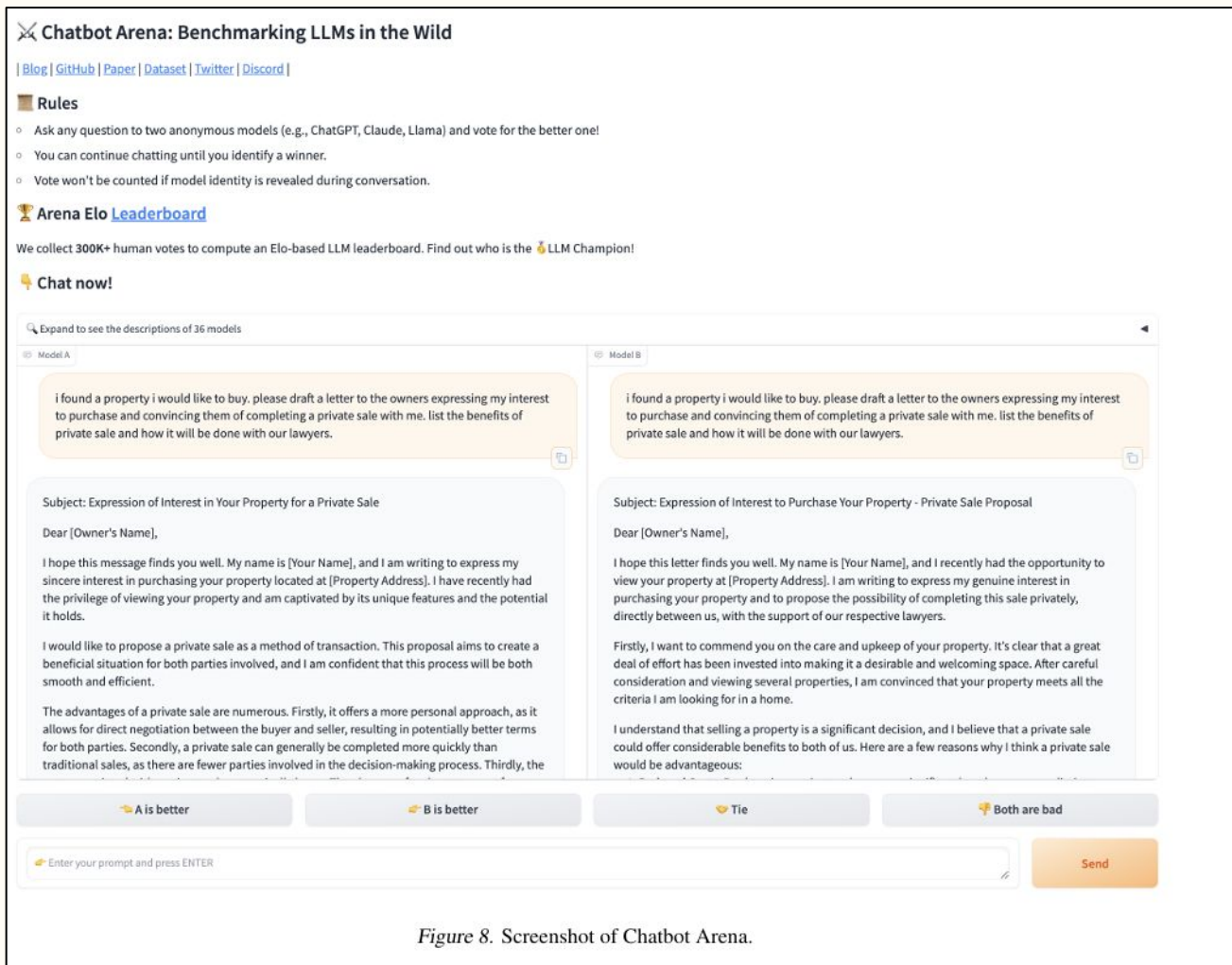


Figure 8. Screenshot of Chatbot Arena.

Why Pairwise Comparison?

Absolute scoring produces inconsistent results across users due to different internal rating scales and calibration standards.

Pairwise judgment is cognitively easier, more reliable across raters, and less sensitive to individual calibration differences.

Humans are better at choosing than scoring.

Why Pairwise Comparison Works - Example

- User A rates Model X: 9/10 and Model Y: 8/10.
- User B rates Model X: 6/10 and Model Y: 5/10.
- Absolute scores differ, but both prefer X over Y.
- Pairwise comparison removes calibration bias and personal rating scale differences.

What Is Elo Rating?

Elo rating is a ranking system originally designed for chess that assigns each model a numerical rating.

When two models are compared, the winner gains rating points while the loser loses points.

Rating updates depend on:

- The expected probability of winning
- The actual outcome

Numerical Example of Elo Update

- Initial ratings: Model A = 1200, Model B = 1200.
- Expected win probability = 0.5.
- If A wins: A increases to ~1216, B decreases to ~1184.
- If lower-rated model wins, rating change is larger.
- Ratings stabilize as more comparisons accumulate.

Why Elo Works for Large Language Model Evaluation

- It accounts for uncertainty in comparisons
- Adjusts dynamically as more votes are collected
- Enables ranking without fixed test sets
- Supports continuous model updates

What Makes Chatbot Arena Different?

Traditional Benchmarks	Chatbot Arena
Fixed datasets	Open-ended prompts
Automatic metrics	Human preference
Static evaluation	Continuous updates
Offline scoring	Live deployment feedback

Chatbot Arena Data Collection

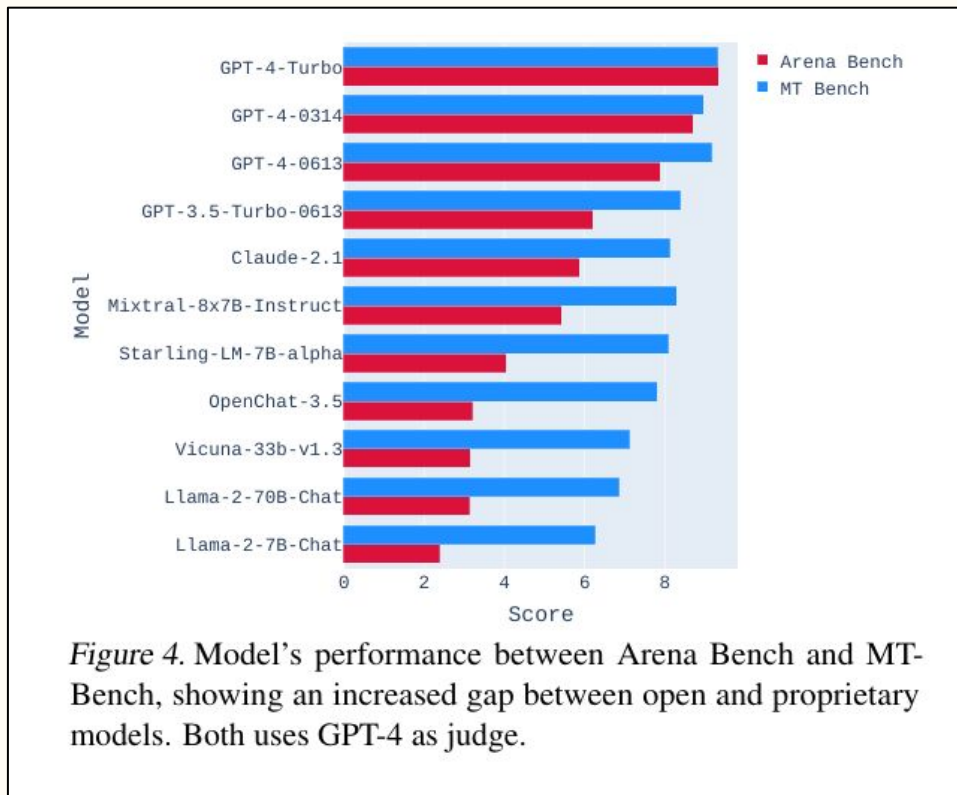
The platform has collected millions of human votes through anonymous blind comparisons, ensuring unbiased evaluation without model identity influencing judgments.

The evaluation covers diverse, real-world prompts spanning multiple domains and task types, reflecting organic user distribution rather than researcher-selected test cases.

Arena Ranking vs Benchmark Performance

A moderate correlation is observed.

Some models rank high on benchmarks but lower in Arena.



Measuring Impact

Leaderboard rankings often diverge from benchmarks, exposing gaps between narrow task scores and real-world utility.

Preference-based evaluations capture qualities like helpfulness, coherence, and instruction-following that metrics miss.

Some models excel on static tests through optimization or contamination but underperform in Arena, highlighting the limits of traditional evaluation.

Inference Mechanics and Deployment Context

Models are evaluated exactly as deployed in production, including their complete decoding strategies and generation parameters.

Latency, output formatting, and interaction style affect user perception and are reflected in preference judgments.

No post-hoc scoring heuristics or controlled conditions. Evaluation captures the full user experience including factors beyond text quality alone, providing authentic assessment of real-world model utility.

Statistical Considerations

Pairwise comparisons assume transitive preferences (if A beats B and B beats C, then A beats C), which may not always hold in practice.

Elo rating assumes independence of matchups, meaning performance on one comparison does not affect others.

Rating stability increases with vote count as more data reduces uncertainty. Early rankings exhibit higher variance and should be interpreted cautiously until sufficient comparisons accumulate.

When Pairwise Comparison Breakdown

- Model A: Concise responses.
- Model B: Detailed explanations.
- Model C: Humorous style.
- Users may prefer $A > B$, $B > C$, but $C > A$.
- This violates transitivity assumption required for Elo ranking.

Preference vs Correctness Limitation

- Prompt: Complex factual or math question.
- Model A: Correct but dry explanation.
- Model B: Incorrect but confident and well-formatted answer.
- Users may prefer Model B despite being wrong.
- Arena measures preference, not reasoning validity.

Limitations

- Sampling bias from non-representative user demographics may skew preference patterns toward specific populations or use cases.
- Prompt distribution skew means certain task types may be overrepresented, affecting generalizability of rankings.
- Gaming or coordinated voting risks could manipulate rankings if users systematically vote for specific models.
- Elo captures preference but not reasoning correctness; models can win votes while producing factually incorrect or logically flawed outputs.

Connection to “Measure of Intelligence”

- ARC measures abstraction and reasoning under controlled priors, providing a principled theoretical framework for assessing generalization efficiency with minimal experience.
- Chatbot Arena measures user-perceived quality at scale through real-world deployment feedback, capturing holistic utility across diverse tasks.
- ARC is principled and theory-driven, focusing on fundamental reasoning capabilities. Arena is empirical and deployment-driven, reflecting practical value in actual use. Neither alone provides complete assessment of model capabilities.

Conclusion

- Static benchmarks are insufficient for modern LLMs
- Human preference provides scalable evaluation signal
- Elo enables dynamic ranking without fixed datasets
- Evaluation must evolve with model scale and deployment

References

Chiang, W., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhang, H., Zhu, B., Jordan, M., Gonzalez, J. E., & Stoica, I. (2024, March 7). Chatbot Arena: an open platform for evaluating LLMs by human preference. arXiv.org.
<https://arxiv.org/abs/2403.04132>



Questions?